

(12) **United States Patent**
Lee

(10) **Patent No.:** **US 12,403,492 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **PUMP CAP**

(56) **References Cited**

(71) Applicant: **SAMHWA CO., LTD**, Uiwang-si (KR)

U.S. PATENT DOCUMENTS

(72) Inventor: **Kyung Chang Lee**, Uiwang-si (KR)

4,732,549 A * 3/1988 von Schuckmann B05B 11/1035
222/207
5,014,881 A * 5/1991 Andris B05B 11/1035
222/207

(73) Assignee: **SAMHWA CO., LTD**, Uiwang-si (KR)

(Continued)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 178 days.

FOREIGN PATENT DOCUMENTS

(21) Appl. No.: **17/913,152**

KR 20-2012-0004976 U 7/2012
KR 10-1963619 B1 3/2019

(Continued)

(22) PCT Filed: **Nov. 13, 2020**

OTHER PUBLICATIONS

(86) PCT No.: **PCT/KR2020/015949**

International Search Report in International Application No. PCT/KR2020/015949, dated Feb. 2, 2021, 3 pages.

§ 371 (c)(1),

(2) Date: **Sep. 20, 2022**

Primary Examiner — Paul R Durand

Assistant Examiner — Randall A Gruby

(87) PCT Pub. No.: **WO2021/206252**

(74) *Attorney, Agent, or Firm* — Park, Kim & Suh, LLC

PCT Pub. Date: **Oct. 14, 2021**

(65) **Prior Publication Data**

US 2023/0173518 A1 Jun. 8, 2023

(30) **Foreign Application Priority Data**

Apr. 7, 2020 (KR) 10-2020-0042160

(51) **Int. Cl.**

B05B 11/10 (2023.01)

(52) **U.S. Cl.**

CPC **B05B 11/1074** (2023.01); **B05B 11/1023** (2023.01); **B05B 11/1047** (2023.01); **B05B 11/1077** (2023.01)

(58) **Field of Classification Search**

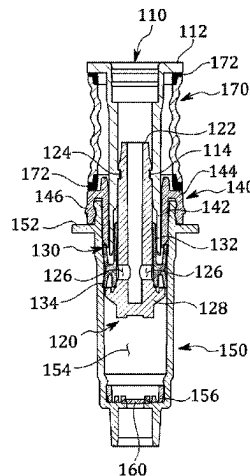
CPC B05B 11/1074; B05B 11/1047; B05B 11/1077; B05B 11/1023

See application file for complete search history.

(57) **ABSTRACT**

One aspect of the invention provides a pump cap that includes: a cylinder having an interior space into which a content may be drawn; a piston configured to move up and down within the interior space and having an inflow hole formed therein that connects to the interior space; a valve configured to move together with the upward and downward movement of the piston, while maintaining contact with an inner perimeter of the cylinder and an outer perimeter of the piston, so as to be capable of closing the inflow hole; a rod that is coupled to an upper portion of the piston, has an inside connecting with the inside of the piston, and protrudes to the outside of the cylinder; a shoulder coupled to an upper portion of the cylinder; and an elastic member that has one end contacting the shoulder and another end contacting the rod and is configured to provide an elastic force for pushing the rod upward, where the elastic member may include a deformable flexible portion.

4 Claims, 5 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

5,238,156	A *	8/1993	Andris	B05B 11/1077	11,773,840	B2 *	10/2023	Ki	B05B 11/1077
				222/207					417/472
5,351,862	A *	10/1994	Weag	B05B 11/028	11,833,533	B2 *	12/2023	Lee	B05B 11/1001
				222/327	12,030,070	B2 *	7/2024	Santagiuliana	B05B 11/1074
5,363,993	A *	11/1994	Mascitelli	B05B 11/1074	12,121,918	B2 *	10/2024	Zembrod	B05B 11/1067
				222/321.9	12,173,768	B2 *	12/2024	Ki	F16F 1/373
5,673,824	A *	10/1997	Evans	B05B 11/1077	2007/0241135	A1 *	10/2007	Santagiuliana	B05B 11/1008
				222/321.9					222/321.9
6,193,112	B1 *	2/2001	Santagiuliana	B05B 11/1074	2009/0206109	A1 *	8/2009	Santagiuliana	B05B 11/1053
				222/321.9					222/145.6
6,672,486	B2 *	1/2004	Santagiuliana	B05B 11/1077	2012/0091229	A1	4/2012	Laidler et al.	
				222/207	2015/0136810	A1	5/2015	Ding	
6,755,327	B1 *	6/2004	Hazard	B05B 11/1059	2017/0326567	A1 *	11/2017	Santagiuliana	B05B 11/1001
				222/321.7	2020/0130002	A1 *	4/2020	Baumann	B05B 11/0044
7,111,761	B2 *	9/2006	Masuda	B05B 11/007	2022/0331824	A1 *	10/2022	Andrew	F16F 9/0418
				222/387	2023/0034207	A1 *	2/2023	Zandona	B05B 11/1035
7,246,723	B2 *	7/2007	Santagiuliana	B05B 11/1059	2023/0074980	A1 *	3/2023	Zembrod	F04B 45/02
				222/207	2023/0204085	A1 *	6/2023	Ki	F16F 1/3605
7,357,278	B2 *	4/2008	Mira Navarro	B05B 11/1077	2024/0066539	A1 *	2/2024	Choe	B05B 11/1023
				222/153.04	2024/0084790	A1 *	3/2024	Byeon	F04B 9/14
9,539,597	B2 *	1/2017	Ding	B05B 11/0027	2025/0035098	A1 *	1/2025	Hang	B05B 11/1077
10,357,792	B2 *	7/2019	Schroeder	B05B 11/1069					
10,421,596	B2 *	9/2019	Koptis	B05C 17/00583					

FOREIGN PATENT DOCUMENTS

KR	10-1975847	B1	5/2019
WO	2009/141510	A1	11/2009
WO	2017/051086	A1	3/2017

* cited by examiner

FIG. 1

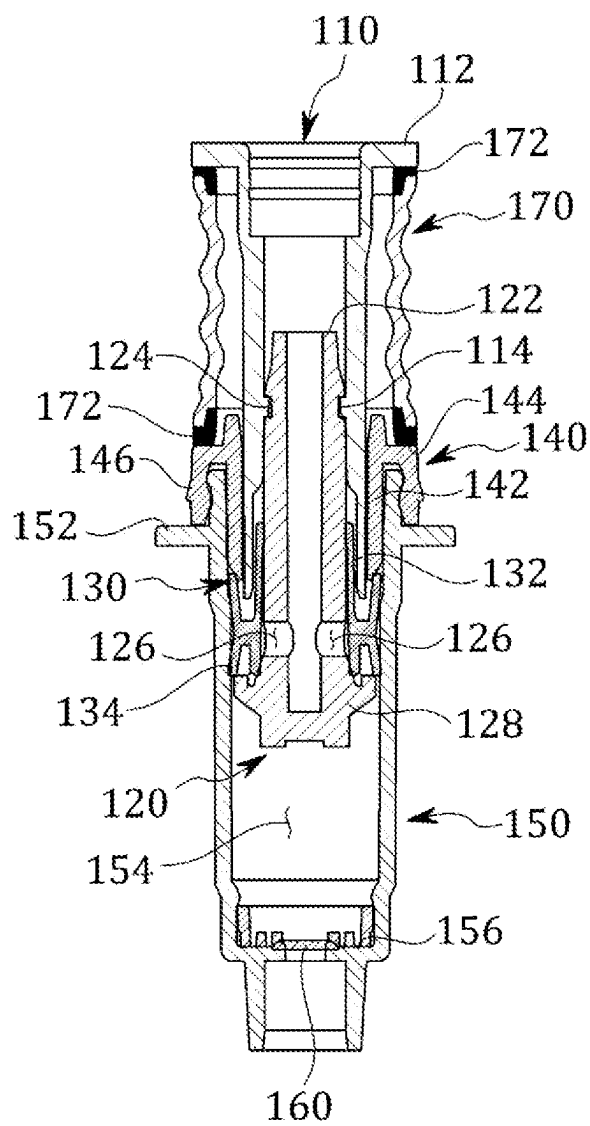


FIG. 2

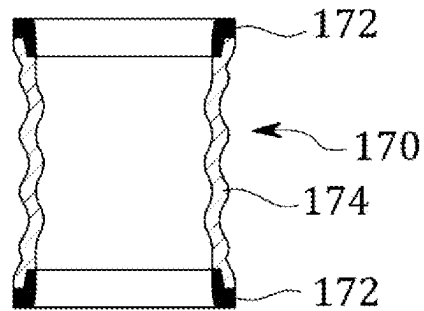


FIG. 3

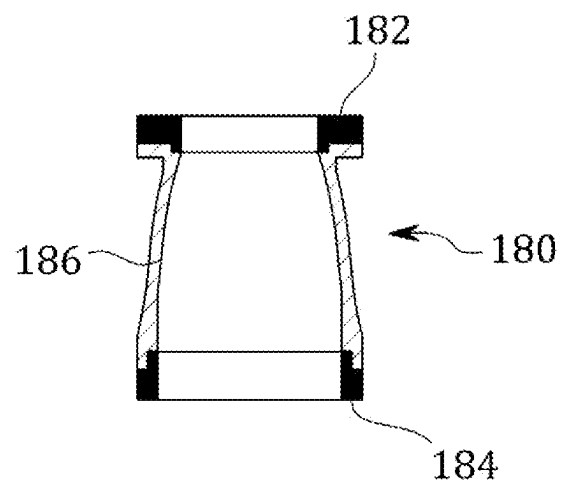


FIG. 4

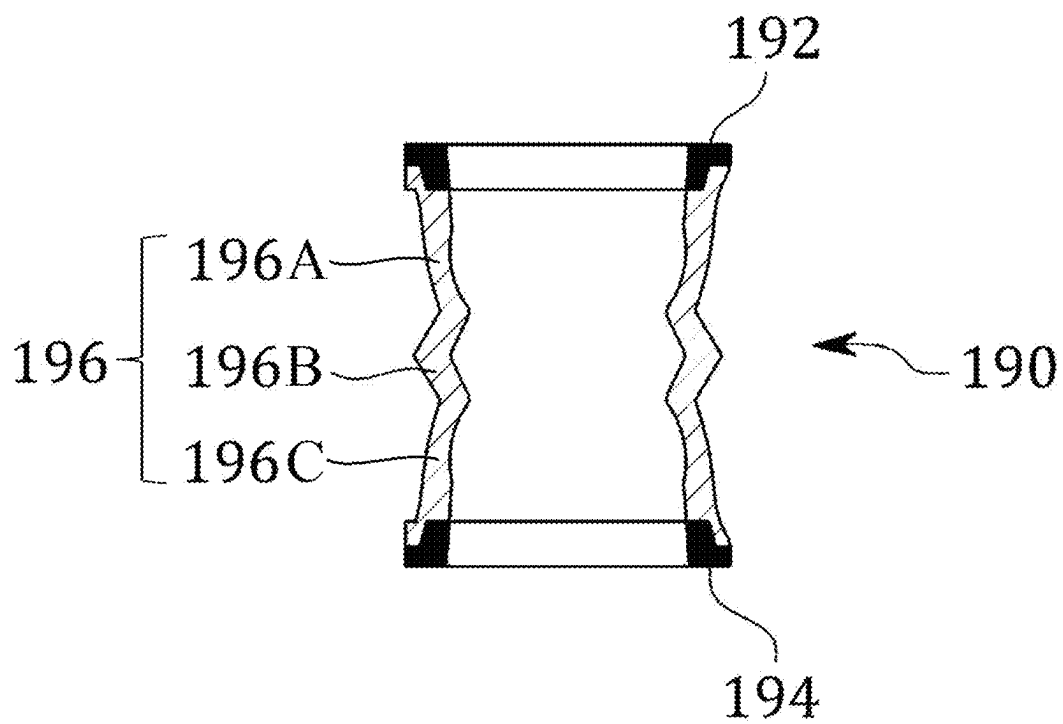


FIG. 5

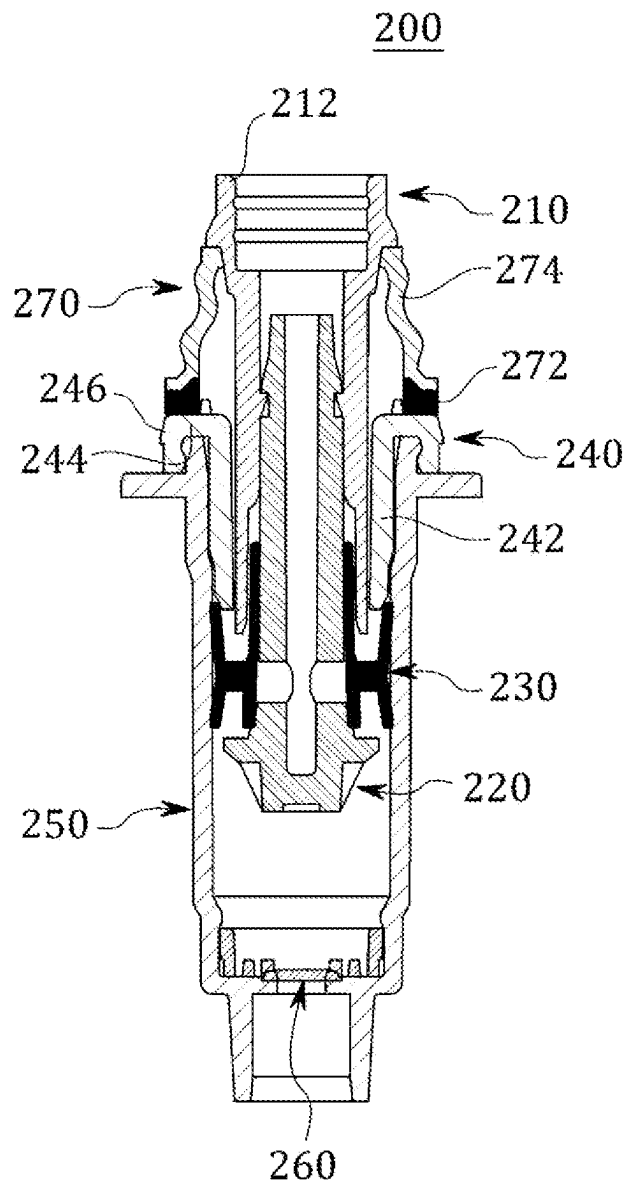
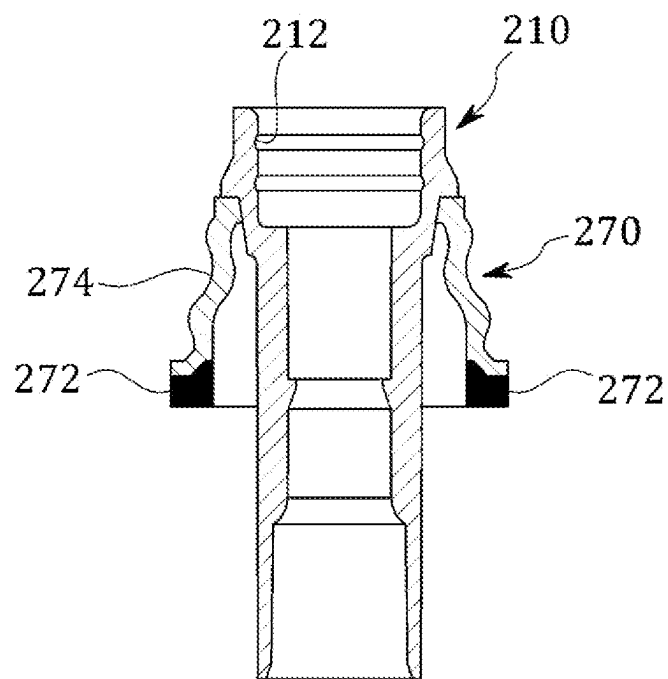


FIG. 6



1

PUMP CAP**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a National Phase Application of PCT International Application No. PCT/KR2020/015949, which was filed on Nov. 13, 2020, and which claims priority from Korean Patent Application No. 10-2020-0042160 filed on Apr. 7, 2020. The disclosures of the above patent applications are incorporated herein by reference in their entirety.

BACKGROUND**1. Technical Field**

The present invention relates to a pump cap that has a simple composition and allows easy recycling.

2. Description of the Related Art

In the context of a cosmetic container, etc., a pump cap may be coupled to the upper opening of a container holding a liquid, such as a cosmetic liquid, so as to dispense a fixed amount of the content to the exterior. A user who wishes to dispense a liquid content may press down on a nozzle corresponding to a button, upon which the content that was drawn into the pump cap may be pressurized, moved up along the discharge path, and subsequently dispensed through the nozzle. When the user removes the pressure on the nozzle, the discharge path may be mechanically closed by the rising of the nozzle, causing a decrease in pressure inside the pump, and the content may be drawn from the container to compensate.

Such a pump cap is being used for dispensing a variety of contents, including not only perfumes and cosmetics but also shampoos, air fresheners, insecticides, and others. In particular, the pump is growing in demand due to its convenient use, as a fixed amount of content can be dispensed by a single pressing of the nozzle, and the content is prevented from leaking to the outside.

Korean Registered Patent No. 1963619 discloses a pump cap for a spray pump. The pump cap disclosed in the above prior art document includes a nozzle, nozzle cap, cap cover, cap, valve, valve spring, housing cover, piston spring, piston, guide, disk, housing, and others, resulting in a complicated composition that is difficult to manufacture. Also, the conventional pump cap includes a valve spring and a piston spring that are made from a metallic material, which is different from the other components and hence makes recycling difficult. In particular, the increase in awareness and need for protecting the environment in recent times have led to a growing demand for a pump cap that is easy to recycle, but the conventional pump cap is as yet unable to satisfy such demand.

SUMMARY OF THE INVENTION

An aspect of the present invention, which was conceived to resolve the problem described above, is to provide a pump cap that has a simple composition and is easy to recycle.

Other objectives of the present invention will be more clearly understood from the embodiments set forth below.

Technical Solution

One aspect of the invention provides a pump cap that includes: a cylinder having an interior space into which a

2

content may be drawn; a piston configured to move up and down within the interior space and having an inflow hole formed therein that connects to the interior space; a valve configured to move together with the upward and downward movement of the piston, while maintaining contact with an inner perimeter of the cylinder and an outer perimeter of the piston, so as to be capable of closing the inflow hole; a rod that is coupled to an upper portion of the piston, has an inside connecting with the inside of the piston, and protrudes to the outside of the cylinder; a shoulder coupled to an upper portion of the cylinder; and an elastic member that has one end contacting the shoulder and another end contacting the rod and is configured to provide an elastic force for pushing the rod upward, where the elastic member may include a deformable flexible portion.

A pump cap according to an embodiment of the present invention can include one or more of the following features. For example, rigid portions can be formed on both ends of the flexible portion.

An upper end of the flexible portion can be formed as an integrated part of the rod, and a rigid portion can be formed on a lower end of the flexible portion. The flexible portion can include a portion that is shaped as a corrugated tube. The portion shaped as a corrugated tube can be formed over the entire length of the flexible portion.

A coupling groove can be formed in an outer perimeter of the piston, and a coupling protrusion configured to be inserted in the coupling groove can be formed on the inner perimeter of the rod.

Advantageous Effects

An embodiment of the present invention having the features above can provide various advantageous effects including the following. However, an embodiment of the present invention may not necessarily exhibit all of the effects below.

An embodiment of the present invention can provide a pump cap that has a simple composition and is easy to recycle.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a cross-sectional view illustrating a pump cap according to a first disclosed embodiment of the invention.

FIG. 2 is a cross-sectional view illustrating the elastic member applied to the pump cap illustrated in FIG. 1.

FIG. 3 and FIG. 4 are cross-sectional views each illustrating an elastic member according to another embodiment of the invention.

FIG. 5 is a cross-sectional view illustrating a pump cap according to a second disclosed embodiment of the invention.

FIG. 6 is a cross-sectional view illustrating the elastic member and rod applied to the pump cap illustrated in FIG. 5.

DETAILED DESCRIPTION OF THE INVENTION

As the invention allows for various changes and numerous embodiments, particular embodiments will be illustrated in the drawings and described in detail in the written description. However, this is not intended to limit the present invention to particular modes of practice, and it is to be appreciated that all changes, equivalents, and substitutes that do not depart from the spirit and technical scope of the

present invention are encompassed by the present invention. In the description of the present invention, certain detailed explanations of the related art are omitted if it is deemed that they may unnecessarily obscure the essence of the invention.

The terms used in the present specification are merely used to describe particular embodiments and are not intended to limit the present invention. An expression used in the singular encompasses the expression of the plural, unless it has a clearly different meaning in the context. In the present specification, it is to be understood that terms such as “including” or “having,” etc., are intended to indicate the existence of the features, numbers, steps, actions, components, parts, or combinations thereof disclosed in the specification and are not intended to preclude the possibility that one or more other features, numbers, steps, actions, components, parts, or combinations thereof may exist or may be added.

While such terms as “first” and “second,” etc., can be used to describe various components, such components are not to be limited by the above terms. The above terms are used only to distinguish one component from another.

Certain embodiments of the present invention will be described below in more detail with reference to the accompanying drawings. Those components that are the same or are in correspondence are rendered the same reference numeral, and redundant descriptions are omitted.

FIG. 1 is a cross-sectional view illustrating a pump cap 100 according to a first disclosed embodiment of the invention, and FIG. 2 is a cross-sectional view illustrating the elastic member 170 applied to the pump cap 100 illustrated in FIG. 1.

Referring to FIG. 1 and FIG. 2, a pump cap 100 according to this embodiment may use an elastic member 170 fabricated from a plastic resin or silicone, etc., instead of the conventional coil spring that is fabricated from a metallic material. The elastic member 170 may have the shape of a corrugated tube, so that when an external force is applied, the elastic member 170 may be elastically compressed due to the contracting of the corrugations, and when the external force is removed, the elastic member 170 may be elastically restored and provide an upward pressure on the rod 110 and piston 120. Forming the elastic member 170 from the same material as that of other parts of the pump cap 100 in this manner can enable easy recycling.

A pump cap 100 based on the first disclosed embodiment may include a rod 110, a piston 120, a valve 130, a shoulder 140, a cylinder 150, a disk 160, and an elastic member 170 and thus can have a simpler composition compared to the conventional pump cap, to therefore provide the advantages of easier fabrication and lowered manufacturing cost.

The rod 110 may be structured to have the shape of a hollow cylinder with a rod flange 112 protruding outward from an upper end thereof. A lower surface of the rod flange 112 may contact the elastic member 170. The top and bottom of the rod 110 may both be open. The button of a pump (not shown) may be coupled to the top of the rod 110, while a portion of the valve 130 and the piston 120 may be inserted into the bottom of the rod 110.

A coupling protrusion 114 may be formed on the inner perimeter of the rod 110. The coupling protrusion 114 may be inserted into a coupling groove 124 formed in the periphery of the piston 120, which is inserted within the rod 110, so that the rod 110 and the piston 120 may move up or down together as an integrated body.

The piston 120 may include a piston body 122 that is inserted into the rod 110. The piston body 122 may be shaped as a hollow cylinder, where the upper end of the

piston body 122 may be open to allow a connection with the interior of the rod 110. Inflow holes 126 may be formed in the periphery of the piston body 122. The valve 130 may be placed in tight contact around the inflow holes 126 so as to close the inflow holes 126.

A piston head 128 may be formed at the lower end of the piston body 122. The piston head 128 may be formed with a larger diameter than that of the piston body 122, so that when the piston 120 moves upward, the valve 130 may be caught on the piston head 128 and be moved upward together.

The valve 130 may be placed in tight contact on the outer perimeter of the piston body 122. The valve 130 may include a piston-contacting portion 132 and a cylinder-contacting portion 134.

The valve 130 may include a piston-contacting portion 132 and a cylinder-contacting portion 134.

The piston-contacting portion 132 may tightly contact the outer perimeter of the piston body 122 and may have the shape of a hollow cylinder. The piston-contacting portion 132 may close the inflow holes 126, but when the piston 120 is moved downward, the piston-contacting portion 132 may move down after a delay following the piston 120 so as to leave the inflow holes 126 open for a certain duration of time. The upper portion of the piston-contacting portion 132 may be inserted within a gap formed between the rod 110 and the piston 120.

The cylinder-contacting portion 134 may have a larger diameter than that of the piston-contacting portion 132 and may be positioned around the piston-contacting portion 132. The cylinder-contacting portion 134 may also have the shape of a hollow cylinder, and the outer perimeter of the cylinder-contacting portion 134 may tightly contact the inner perimeter of the cylinder 150. Due to the cylinder-contacting portion 134, the valve 130 may be delayed from moving downward when the piston 120 is moved down.

Thus, the valve 130 may be delayed in its downward movement, due to the friction with the inner perimeter of the cylinder 150, when the piston 120 moves downward. As a result, the inflow holes 126 of the piston 120 may be opened, and the content (not shown) that had been drawn into the interior space 154 may be drawn into the inside of the piston 120. When the piston 120 moves upward, the valve 130 may be caught on the piston head 128 and may move up together with the piston 120.

The shoulder 140 may include a shoulder body 142 that is inserted into the open top of the cylinder 150. The shoulder body 142 may have the shape of a hollow cylinder and may receive the rod 110 inserted therein. A shoulder flange 144 may protrude outward from a periphery of the shoulder body 142. The upper surface of the shoulder flange 144 may contact the lower end of the elastic member 170. Also, a shoulder periphery part 146 may be formed protruding downward from the end of the shoulder flange 144. The shoulder periphery part 146 may contact the outer perimeter of an upper portion of the, and the lower end of the shoulder periphery part 146 may contact an upper surface of the cylinder flange 152.

The cylinder 150 may be inserted into the opening in an upper portion of the container (not shown) in which the content is held. The cylinder 150 may include an interior space 154 into which the content may be drawn, where the piston 120 and the valve 130 may be located in and may undergo upward and downward movement within the interior space 154.

The cylinder 150 may have the shape of a hollow cylinder and may be structured such that the top and the bottom are

5

both open. A cylinder flange **152** may protrude outward from a periphery of the cylinder **150**. The cylinder flange **152** can be mounted on the upper portion of the container (not shown).

On the inside of the cylinder **150**, a mounting curb **156** having a reduced diameter may be formed. The disk **160** can be positioned on the mounting curb **156**. The disk **160** can be configured to open or close depending on the pressure difference between the interior space **154** and the container. A description of the structure of the disk **160** is not provided here, as it is readily available in documents such as Korean Registered Patent No. 1975847.

The elastic member **170**, while positioned between the rod flange **112** and the shoulder flange **144**, may serve to push the rod **110** upward. When an external force is applied, the elastic member **170** may be elastically deformed, as the flexible portion **174** is compressed.

The elastic member **170** may include a flexible portion **174**, which is capable of elastic deformation, and rigid portions **172**, each of which may be formed on either end of the flexible portion **174**. The flexible portion **174** and the rigid portions **172** can be formed as an integrated body by way of double-shot molding. Also, the flexible portion **174** and the rigid portions **172** can be formed from a material such as plastic resin, etc., that is the same as the material used for other parts of the pump cap **100**.

The rigid portion **172** located at the lower end of the elastic member **170** may be positioned on the upper surface of the shoulder flange **144**. The rigid portion **172** located at the upper end of the elastic member **170** may be positioned at the lower surface of the rod flange **112**. The rigid portions **172** may be formed by a material having a certain level of hardness, so as to support both ends of the elastic member **170** in a stable manner.

The flexible portion **174** may be shaped as a corrugated tube that is capable of being compressed. The flexible portion **174** may have the shape of a hollow cylinder, and a portion of the rod **110** may be positioned on the inside of the flexible portion **174**.

Although an example of an elastic member **170** according to the first disclosed embodiment is illustrated with the flexible portion **174** having the same diameter along its height and having the shape of a corrugated tube over its entire length, an embodiment of the invention can include an elastic member **180**, **190** having any of a variety of shapes, as is illustrated in FIG. 3 and FIG. 4.

FIG. 3 and FIG. 4 are cross-sectional views each illustrating an elastic member **180**, **190** according to another embodiment of the invention.

Referring to FIG. 3, an elastic member **180** can include two rigid portions **182**, **184** that are shaped differently. Also, the flexible portion **186**, which is capable of being compressed and providing an elastic force, can have a smooth shape instead of being shaped as a corrugated tube. When pressed, a flexible portion **186** shaped in this manner can be elastically deformed as a portion thereof is folded.

Referring to FIG. 4, an elastic member **190** can include two rigid portions **192**, **194** that are shaped alike. Also, a flexible portion **196** that is capable of being compressed and providing an elastic force may be formed between the rigid portions **192**, **194**. The flexible portion **196** may have a corrugation in only a portion at the center, rather than having corrugations along its entire length. The flexible portion **196** includes an upper non-corrugated tube **196A**, a lower non-corrugated tube **196C**, and a center corrugated tube **196B** provided between the upper and lower non-corrugated tubes **196A** and **196C**.

6

Thus, embodiments of the invention can include elastic members having various shapes and structures.

FIG. 5 is a cross-sectional view illustrating a pump cap **200** according to a second disclosed embodiment of the invention, and FIG. 6 is a cross-sectional view illustrating the elastic member **270** and rod **210** applied to the pump cap **200** illustrated in FIG. 5.

Referring to FIG. 5 and FIG. 6, a pump cap **200** based on the second disclosed embodiment may have the same or similar components as those of a pump cap **100** based on the first disclosed embodiment, and as such, the following descriptions will focus mainly on the differences.

A pump cap **200** according to the second embodiment may include a rod **210**, a piston **220**, a valve **230**, a shoulder **240**, a cylinder **250**, a disk **260**, and an elastic member **270**. Among these, the piston **220**, valve **230**, cylinder **250**, and disk **260** may be substantially the same as their counterparts in a pump cap **100** according to the first disclosed embodiment and thus will not be described here in detail.

The rod **210** may be substantially the same as the rod **110** in the first disclosed embodiment, but with differences in the shape of the rod head **212** and the fact that the upper end of the flexible portion **274** may be formed as an integrated part on the periphery of the rod head **212**.

The shoulder **240** may also be substantially the same as the shoulder **140** in the first disclosed embodiment, but with the difference that the shoulder flange **244** formed on the periphery of the shoulder body **242** may be formed on the upper end. The shoulder periphery part **246** may also be substantially the same as the shoulder periphery part **146** of the first disclosed embodiment.

An elastic member **270** based on the second disclosed embodiment may not have a rigid portion formed on the upper end of the flexible portion **274** but rather may be formed as an integrated body with the rod head **212**. Also, the flexible portion **274** may be shaped as a corrugated tube has a decreasing diameter from the bottom upward. A rigid portion **272** that is harder and less easily deformed than the flexible portion **274** may be formed at the lower end of the flexible portion **274**. The rigid portion **272** may be positioned on the upper surface of the shoulder flange **244**.

While the foregoing provides a description with reference to an embodiment of the present invention, it should be appreciated that a person having ordinary skill in the relevant field of art would be able to make various modifications and alterations to the present invention without departing from the spirit and scope of the present invention set forth in the scope of claims below.

What is claimed is:

1. A pump cap comprising:

- a cylinder having an interior space configured to receive a content drawn thereinto;
- a piston configured to move up and down within the interior space and having an inflow hole formed therein, the inflow hole connecting to the interior space;
- a valve configured to move together with an upward and downward movement of the piston while maintaining contact with an inner perimeter of the cylinder and an outer perimeter of the piston, the valve capable of closing the inflow hole;
- a rod coupled to an upper portion of the piston and having an inside thereof connecting with an inside of the piston, the rod protruding to an outside of the cylinder;
- a shoulder coupled to an upper portion of the cylinder; and
- an elastic member having one end thereof contacting the shoulder and another end thereof contacting the rod, the

- elastic member configured to provide an elastic force
for pushing the rod upward,
wherein the elastic member comprises a flexible portion,
wherein the flexible portion comprises:
 an upper non-corrugated tube; 5
 a lower non-corrugated tube; and
 a center corrugated tube provided between the upper
 and lower non-corrugated tubes,
wherein an outer diameter of an upper end of the upper
non-corrugated tube is greater than an outer diameter of 10
a lower end of the upper non-corrugated tube,
wherein an outer diameter of an upper end of the lower
non-corrugated tube is less than an outer diameter of a
lower end of the lower non-corrugated tube,
wherein a length of the center corrugated tube is less than 15
lengths of the upper and lower non-corrugated tubes,
respectively.
2. The pump cap of claim 1, wherein rigid portions are
formed on both ends of the flexible portion.
3. The pump cap of claim 1, wherein an upper end of the 20
flexible portion is formed as an integrated part of the rod,
and a rigid portion is formed on a lower end of the flexible
portion.
4. The pump cap of claim 1, wherein a coupling groove
is formed in an outer perimeter of the piston, and a coupling 25
protrusion configured to be inserted in the coupling groove
is formed on an inner perimeter of the rod.

* * * * *